Rupak Ghosh

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Begumpur, Hooghly, West Bengal **Date of Birth:** 30/09/2005

Summary

Computer Science (AI & ML) undergraduate with a **9.5/10 CGPA**, focused on **C, Data Structures & Algorithms, and problem-solving** for Software Engineering roles. Skilled in Python, SQL, Machine Learning, Generative AI, and web development, with hands-on experience building practical applications. Aspiring to build a career as a **Software Engineer** and develop efficient, scalable solutions.

Education

Brainware University, Barasat **2023 – 2027**
B.Tech in Computer Science & Engineering (AI & ML) *CGPA: 9.5 / 10*

Begumpur High School, West Bengal **2023**
Higher Secondary Education *90.2%*

Baratajpur High School, West Bengal **2021**
Secondary Education *87.57%*

Skills

Programming: C, Python, CPP

Data Structures & Algorithms: Arrays, Linked Lists, Stacks, Queues, Trees, Searching, Sorting

Core CS: OOP, DBMS, SQL

Machine Learning: Supervised Learning, Unsupervised Learning, Feature Engineering, Model Evaluation, Linear Regression, Logistic Regression, KNN, SVM, Decision Trees, Random Forest, Clustering

Data Analysis & Visualization: Data Analysis Using Python, Data Preprocessing, Exploratory Data Analysis (EDA), Data Visualization, Power BI

Deep Learning: Neural Networks, ANN, CNN, Deep Learning Fundamentals

Python Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Keras

Web Development: HTML5, CSS3, JavaScript, React.js, Tailwind CSS, Node.js

Tools: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Postman

Projects

Human Emotion Detection using CNN | Python, TensorFlow/Keras, CNN, OpenCV **Sep 2026**

- Built and trained a **CNN-based deep learning model** for facial emotion recognition across **7 classes**: Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral.
- Performed image preprocessing including **resizing, normalization, and data augmentation** to improve model training and generalization.
- Integrated **OpenCV face detection** for real-time facial image processing and webcam-based emotion prediction.
- Analyzed model performance using **accuracy, precision, recall, F1-score, and confusion matrix**.

Smart Data Cleaner [Live Demo] | Flask, PostgreSQL, Pandas, Plotly **Jul 2026 – Aug 2026**

- Built a web-based platform for **automated data cleaning, quality scoring, and CSV/Excel analysis** using Flask and Pandas.
- Implemented **authentication, cleaning history, missing-value handling, and interactive Plotly visualizations** with PostgreSQL integration.

Spotify Clone Backend | Node.js, Express.js, MongoDB **Feb 2026 – Mar 2026**

- Developed **RESTful APIs** with authentication, database schemas, middleware, and backend routing.

Credit Card Fraud Detection | Python, Scikit-learn **Jul 2025 – Aug 2025**

– Built fraud detection models for **imbalanced datasets** and evaluated performance using precision, recall, and F1-score.

Fake News Detection | Python, Scikit-learn, NLP **Jun 2025**

– Developed an NLP-based classifier to detect **Fake and Real news** using TF-IDF and Logistic Regression.

– Performed text preprocessing and feature extraction to convert news articles into machine-readable data.

– Evaluated model performance using **accuracy, precision, recall, F1-score, and confusion matrix**.

Certifications & Achievements

- **TCS iON NQT-IT 2026** – **60.64% Overall**(2026)
- **NCET 2026** – **67.50%**, Grade A on first attempt.(2026)
- **1st Place** – Textibition Technical Exhibition (2025)
- **2nd Place** – Code for Change, NSHM College – *Robotic Arm Car* (2026)
- **Artificial Intelligence Training** – Samsung Innovation Campus (2025)
- **Career Essentials in Generative AI** – Microsoft & LinkedIn (2025)
- **Generative AI for All** – Infosys Springboard (2025)

Date: _____ **Signature:** _____

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